

Project Report

Meeting Room Booking Platform

Abstract:

The Meeting Room Booking System is a web-based application designed to simplify and automate the process of reserving meeting rooms within an organisation. Using a MySQL backend and PHP for server-side processing, the system allows employees to book rooms, select available options, and calculate fare based on hourly room rates. This project demonstrates the use of a relational database with multiple tables, SQL queries, and CRUD operations. The system improves efficiency, reduces scheduling conflicts, and provides a structured way to manage internal meeting room usage.

Chapter 1: Introduction

Efficient allocation of meeting rooms is essential in modern organisations, especially where multiple teams need collaborative spaces. Traditional manual booking methods cause confusion, double-booking, and poor resource utilisation. To address these issues, a digitised booking system helps streamline the reservation workflow.

The Meeting Room Booking System provides employees with a user-friendly interface to enter details, choose rooms, and receive fare calculation based on hourly rates. The backend stores employee data, room details, and booking transactions, ensuring accuracy and traceability.

Chapter 2: Literature Review

Paper 1: Studies on resource management systems highlight that database-driven applications reduce administrative overhead and improve data consistency. Such systems ensure real-time availability tracking and structured record maintenance.

Paper 2: Research on room reservation systems suggests automation reduces scheduling conflicts by up to 60% while improving user satisfaction through self-service interfaces.

Paper 3: Literature on web-based booking platforms emphasises the importance of data integrity, secure transactions, and usability for smooth organisational workflows.

Paper 4: Academic works on DBMS applications show that multi-table relational models improve scalability and adaptability for future extension, such as analytics or audit logs.

Paper 5: Studies on cost-calculation modules in booking systems demonstrate how automated pricing eliminates human error and ensures transparent billing.

2.1 Problem Statement

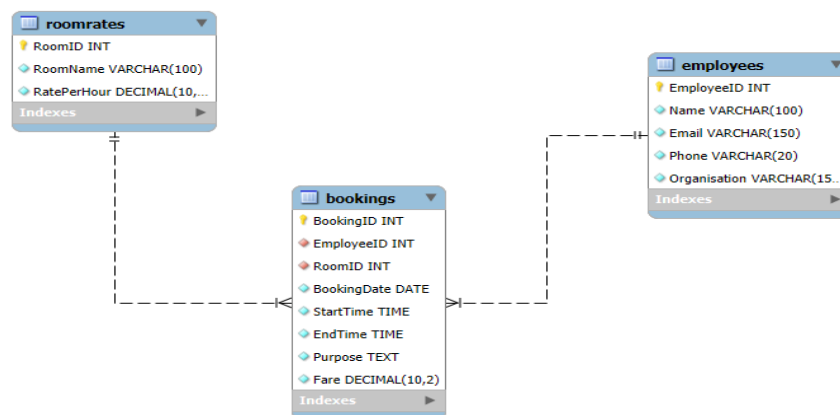
Organisations face difficulties in managing meeting room availability due to manual processes, miscommunication, and lack of centralised records.

2.2 Objectives

- To design a database-driven system for booking meeting rooms.
- To calculate fare based on hourly room rates.
- To store employee and booking details securely.
- To prevent double-booking or data inconsistency.
- To provide a user-friendly interface for seamless booking.

Chapter 3: Methodology

Database Schema Diagram:



Architecture Diagram:

1. **Presentation Layer (Frontend):** HTML, CSS, JavaScript.
2. **Application Layer (Backend):** PHP scripts handling logic, validation, and calculations.
3. **Data Layer (Database):** MySQL database storing persistent data (XAMPP).

Flow:

- User Interface → PHP Backend → MySQL Database (XAMPP) → Response to UI

3.1 Functional Requirements

1. The system shall allow users to register and log in securely.
2. The system shall enable employees to book meeting rooms by selecting date and time.
3. The system shall calculate the booking fare based on duration and room rate per hour.
4. The system shall store and retrieve booking details from the database.
5. The system shall store and retrieve booking details from the database.

3.2 Non-Functional Requirements

1. **Performance:** The system should respond to user requests within 2 seconds.
 2. **Usability:** User authentication and session management should prevent unauthorized access.
-

Chapter 4: Results

Explanation:

- **Authentication Module:** Users can register and log in using validated credentials.
- **Dashboard Page:** Displays all bookings made by the logged-in user with options to edit or delete.
- **Booking Form:** Employee enters name, email, date, time, and purpose.
- **Room Selection Page:** User selects one of the available meeting rooms.
- **Summary Page:** Shows all booking details, fare, and confirmation.
- **Invoice Generation:** A downloadable invoice is generated after successful booking.
- **Database:** Tables created — employees, roomrates, bookings.
- **PHP Backend:** Successfully calculates fare and stores booking records.

Screenshot:

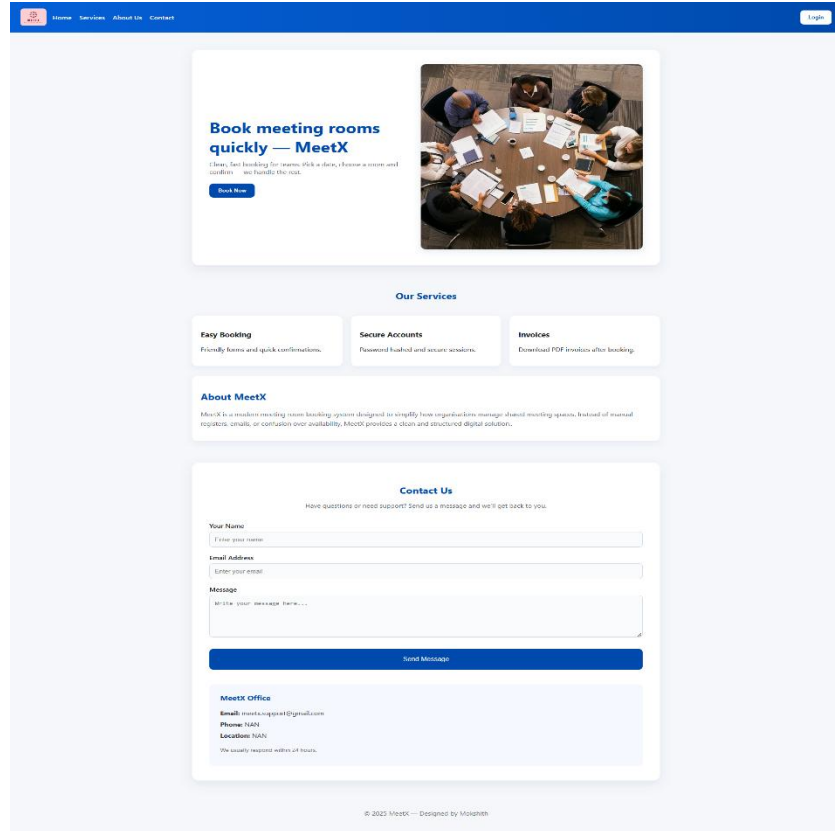


Figure 1: Home page

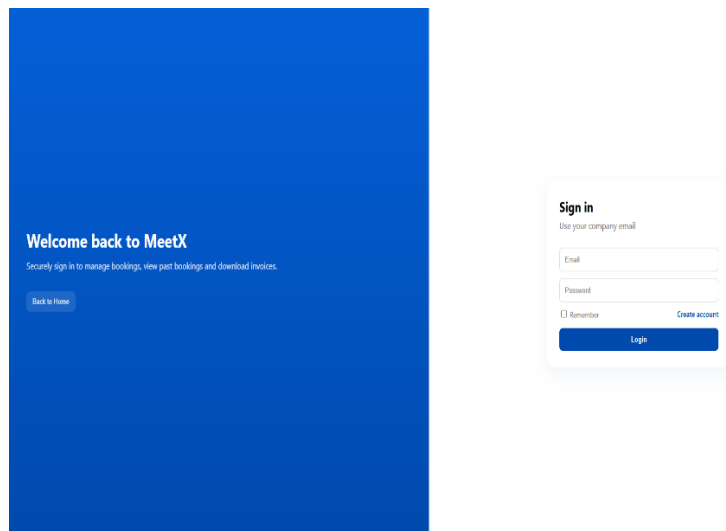
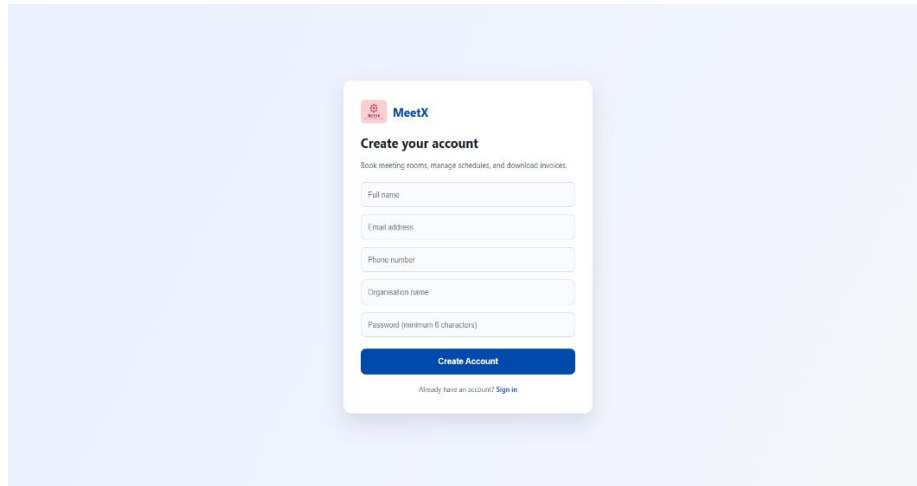
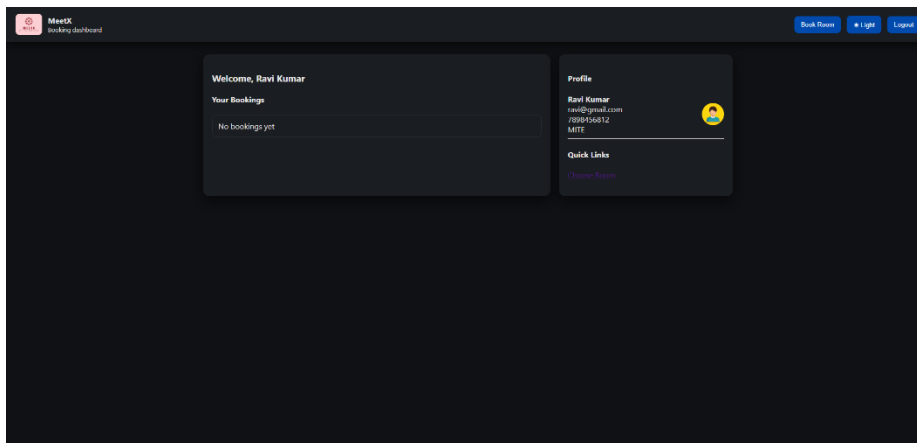


Figure 2: Login page



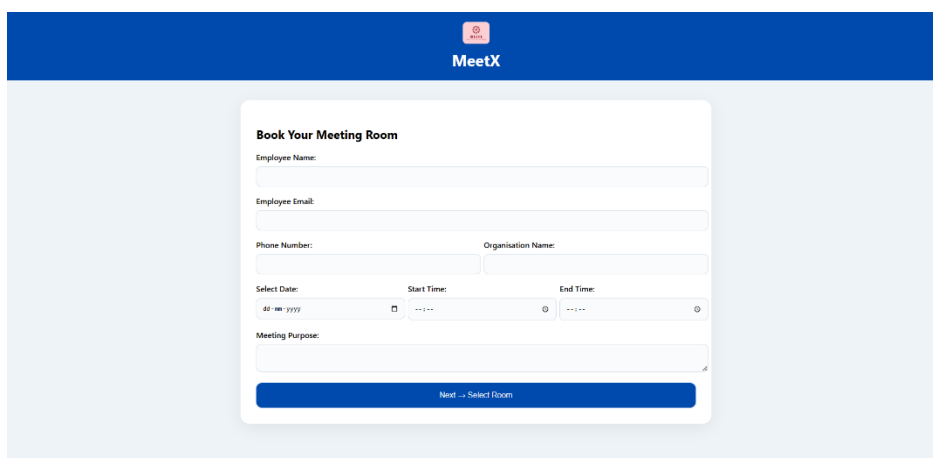
The registration form is titled "MeetX" and "Create your account". It includes a sub-header: "Book meeting rooms, manage schedules, and download invoices." The form contains the following fields: Full name, Email address, Phone number, Organisation name, and Password (minimum 6 characters). A "Create Account" button is at the bottom. A link "Already have an account? Sign in" is located below the password field.

Figure 3: Register



The dashboard is titled "MeetX booking dashboard". It features a dark theme with a top navigation bar containing "Book Room", "Light", and "Logout" buttons. The main content area is divided into two sections: "Welcome, Ravi Kumar" and "Your Bookings". The "Your Bookings" section shows "No bookings yet". The "Profile" section displays the user's name "Ravi Kumar", email "ravi@gmail.com", phone number "9886154812", and company "MITT". A "Quick Links" section with a "Choose Room" button is also present.

Figure 4: Dashboard



The booking form is titled "MeetX" and "Book Your Meeting Room". It includes the following fields: Employee Name, Employee Email, Phone Number, and Organisation Name. There are also fields for "Select Date" (format dd-mm-yyyy), "Start Time" (format --:--), and "End Time" (format --:--). A "Meeting Purpose" field is at the bottom. A "Next -> Select Room" button is located at the bottom of the form.

Figure 5: Booking



Select Your Meeting Room

Auditorium
Capacity: 500 people
₹50000.00/hr

Conference Room A
Capacity: 14 people
₹20000.00/hr

Conference Room B
Capacity: 10 people
₹18000.00/hr

Meeting Room A
Capacity: 5 people
₹12000.00/hr

Meeting Room B
Capacity: 5 people
₹12000.00/hr

Project Room
Capacity: 3 people
₹15000.00/hr

Confirm →

© 2025 MeetX

Figure 6: Room



Review Booking

Name: Ravi Kumar

Email: ravi@gmail.com

Phone: 7898456812

Organisation: MITE

Booking Details

Date: 2026-11-02

Start Time: 12:00

End Time: 10:00

Purpose: Tech

Room: Project Room

Fare

Total Fare: ₹330000 (22 hr at ₹15000/hr)

BOOK NOW

Figure 7: Summary



MeetX

Booking Invoice

Booking Information	
Booking ID	7
Date	2026-11-02
Start Time	12:00:00
End Time	10:00:00
Purpose	Tech
Room	Project Room
Rate Per Hour	₹15000.00
Employee Information	
Name	Ravi Kumar
Email	ravi@gmail.com
Phone	7898456812
Organisation	MITE
Total Fare: ₹30000.00	

Figure 8: Invoice

Code Snippet:

```
1 CREATE DATABASE IF NOT EXISTS meetingroomsys;
2 USE meetingroomsys;
3
4 CREATE TABLE employees (
5     EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
6     Name VARCHAR(100) NOT NULL,
7     Email VARCHAR(150) NOT NULL,
8     Phone VARCHAR(20) NOT NULL,
9     Organisation VARCHAR(150) NOT NULL
10 );
11
12
13 CREATE TABLE roomrates (
14     RoomID INT AUTO_INCREMENT PRIMARY KEY,
15     RoomName VARCHAR(100) NOT NULL,
16     RatePerHour DECIMAL(10,2) NOT NULL
17 );
18
19 INSERT INTO roomRates (RoomName, RatePerHour) VALUES
20 ('Conference Room A', 20000),
21 ('Conference Room B', 25000),
22 ('Project Room', 15000),
23 ('Meeting Room A', 18000),
24 ('Meeting Room B', 17000),
25 ('Auditorium', 70000);
26
27 CREATE TABLE bookings (
28     BookingID INT AUTO_INCREMENT PRIMARY KEY,
29     EmployeeID INT NOT NULL,
30     RoomID INT NOT NULL,
31     BookingDate DATE NOT NULL,
32     StartTime TIME NOT NULL,
33     EndTime TIME NOT NULL,
34     Purpose TEXT NOT NULL,
35     Fare DECIMAL(10,2) NOT NULL,
36     FOREIGN KEY (EmployeeID) REFERENCES employees(EmployeeID),
37     FOREIGN KEY (RoomID) REFERENCES roomRates(RoomID)
38 );
```

Chapter 5: Conclusion

The Meeting Room Booking System successfully demonstrates a complete DBMS-based web application using a relational database approach. The project integrates frontend interfaces with a PHP backend and a MySQL database to provide secure authentication, automated fare calculation, and efficient booking management.

The system reduces manual effort, prevents scheduling conflicts, and improves organizational efficiency. Future enhancements may include real-time room availability tracking, email notifications, and role-based access control.

References

1. "Web-based Resource Management Systems", Journal of Computing, 2021.
2. "Database Design for Reservation Systems", IEEE Research, 2020.
3. "Automation in Organisational Tools", ACM Digital Library, 2022.
4. "SQL-based Room Booking Applications", International Journal of Computer Science, 2019.
5. "Cost Calculation Models in Booking Systems", IJERT, 2021.